

MASTER IN DATA SCIENCE

Complex and Social Networks Laboratory

Finding and Assessing Community Structure

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Laboratory 6 Report

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1 Introduction

In this laboratory, we analyzed community structures in various complex networks, aiming to compare the performance of different community detection algorithms and evaluate their correspondence with reference partitions.

To this end, we implemented an R pipeline to compute cluster similarity using the Jaccard index. The function `jaccard_sim()` measures similarity between individual clusters, `match_clusters()` identifies the most similar clusters with respect to a reference partition, and `Wmean()` provides a global similarity score weighted by cluster sizes.

We applied four community detection algorithms: Louvain, Label Propagation, Walktrap, and Edge Betweenness on four different networks:

- the Karate Club
- a synthetic scale-free model
- the ENRON email network
- the Macaque cortical connectivity network

For each network, we calculated significance measures such as modularity, conductance, and clustering coefficient, as well as local and global Jaccard-based similarities with the reference partition, either a ground truth or the best-scoring solution based on averaged significance scores.

This analysis provides a comprehensive comparison of the algorithms, highlighting their strengths, limitations, and consistency with known or inferred community structures.

2 Results

2.1 Karate Network

The community detection results for the Karate Club network are summarized in Tables 1 through 4 and visualized in Figure 1.

In this case, the Ground Truth (the actual faction split observed by Zachary) was used as the reference partition. The significance scores for this ground truth and the detected partitions are presented in Table 2. It is notable that while the Ground Truth has the lowest Conductance (0.0952), algorithms like Walktrap ($Q \approx 0.4112$) and Louvain ($Q \approx 0.4088$) actually achieve higher modularity scores than the real-world division ($Q \approx 0.3715$).

Table 1 presents the Global Weighted Mean Jaccard Scores (J_{global}). Label Propagation achieved the highest similarity to the ground truth ($J_{global} = 0.7163$). Walktrap followed closely with 0.7059, while Edge Betweenness showed the lowest fidelity (0.5588).

The granular breakdown of these similarities is detailed in Table 3 (Matched Clusters) and Table 4 (Jaccard Similarity Matrices).

- **Label Propagation (Table 3b, 4b):** This algorithm demonstrated robust recovery of the two factions. It matched reference cluster C1.2 (the "Officers" faction) with high accuracy (0.7778) to C2.3, and C1.1 (the "Mr. Hi" faction) to C2.1 with a score of 0.6471.
- **Louvain and Walktrap (Table 3a/c, 4a/c):** Both modularity-based methods showed a tendency to subdivide the main factions. For instance, Louvain split the ground truth cluster C1.1 into two subgroups (C2.1 and C2.2), reducing the primary match score to 0.6875.
- **Edge Betweenness (Table 3d, 4d):** This method produced the most fragmented partition, splitting the network into 6 smaller communities (C2.1 through C2.6). Consequently, the best match for the reference cluster C1.1 only reached a score of 0.5625, significantly lower than the other methods.

Finally, Figure 1 illustrates the topological layout of these communities. A detailed interpretation of why modularity optimization yields finer partitions than the sociological ground truth will be provided in the Discussion section.

Table 1: Global and Weighted Jaccard Scores for Karate Club

	J_{global}	J_w
Louvain	0.6765	0.6771
Label Propagation	0.7163	0.7124
Walktrap	0.7059	0.7049
Edge Betweenness	0.5588	0.5590

Table 2: Significance scores for Karate Club

	Louvain	Label Prop.	Walktrap	EdgeBetw	Ground Truth
Modularity	0.4088	0.3323	0.4112	0.3619	0.3715
Conductance	0.2306	0.1687	0.2548	0.4194	0.0952
Clustering Coef	0.6025	0.5101	0.6138	0.4505	0.5341
VI dist to GT	0.7835	0.8273	0.8729	1.6382	0.0000

Table 3: Matched Clusters and Local Jaccard Scores (Reference: Ground Truth)

a) Louvain vs Ground Truth			b) Label Propagation vs Ground Truth		
Ref (GT)	Match	Score	Ref (GT)	Match	Score
C1.1	C2.1	0.6875	C1.1	C2.1	0.6471
C1.2	C2.3	0.6667	C1.2	C2.3	0.7778

c) Walktrap vs Ground Truth			d) Edge Betweenness vs Ground Truth		
Ref (GT)	Match	Score	Ref (GT)	Match	Score
C1.1	C2.2	0.6875	C1.1	C2.1	0.5625
C1.2	C2.1	0.7222	C1.2	C2.4	0.5556

Table 4: Jaccard Similarity Matrices for Karate Club (Reference: Ground Truth)

a) Louvain					b) Label Propagation				
	C2.1	C2.2	C2.3	C2.4		C2.1	C2.2	C2.3	C2.4
C1.1	0.6875	0.3125	0.0000	0.0000	C1.1	0.6471	0.3125	0.0000	0.0000
C1.2	0.0000	0.0000	0.6667	0.3333	C1.2	0.0345	0.0000	0.7778	0.1667

c) Walktrap					d) Edge Betweenness						
	C2.1	C2.2	C2.3	C2.4		C2.1	C2.2	C2.3	C2.4	C2.5	C2.6
C1.1	0.0000	0.6875	0.3125	0.0000	C1.1	0.5625	0.1111	0.3125	0.0000	0.0000	0.0000
C1.2	0.7222	0.0000	0.0000	0.2778	C1.2	0.0000	0.1000	0.0000	0.5556	0.2222	0.1111

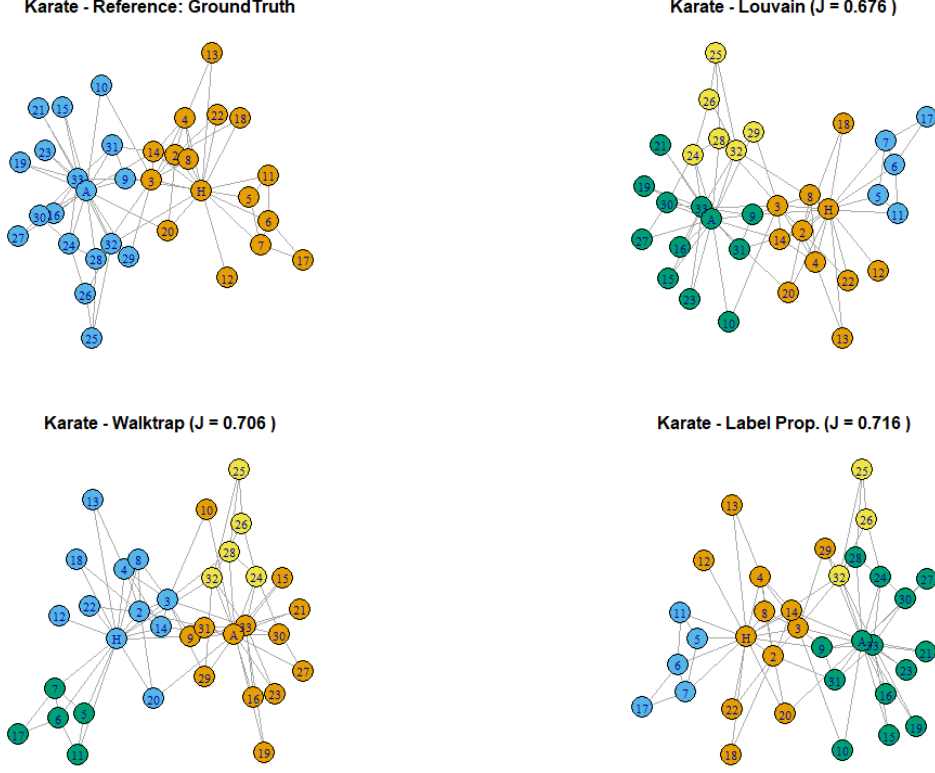


Figure 1: Karate club network.

2.2 Synthetic Network

The community detection results for the Synthetic network are summarized in Tables 6 and visualized in Figure 2.

The Ground Truth (the planted partition of 4 communities) was used as the reference. The significance scores in Table 6 reveal a critical insight: the Ground Truth partition itself possesses a relatively low modularity ($Q \approx 0.1666$) and high conductance (0.5748). Interestingly, both Louvain ($Q \approx 0.2958$) and Walktrap ($Q \approx 0.2213$) identified partitions with higher modularity than the planted ground truth.

Table 5 presents the Jaccard scores. A stark contrast is observed here compared to previous networks: the Weighted Jaccard scores (J_w) are universally low, ranging from 0.15 to 0.25. Tables 8 and 9 report the same type of analysis; however, Table 9 focuses exclusively on the results obtained with the Edge Betweenness algorithm, isolating its Jaccard similarity scores for clarity and comparison. The granular breakdown in Tables 7, 8 and 9 clarifies the nature of this failure:

- **Label Propagation (Table 7b, 8b):** This algorithm failed to detect the community structure entirely, assigning all four reference clusters to a single detected cluster (C2.1). This trivial partition results in match scores of approximately 0.20 – 0.33, which roughly corresponds to the ratio of a single community size to the total network size.

- **Louvain and Walktrap (Table 7a/c, 8a/c):** Unlike Label Propagation, these methods did partition the network, but the resulting clusters show little correspondence to the ground truth. The similarity matrices are dense with low values. For example, Louvain’s best match for reference cluster C1.1 is C2.8 with a score of only 0.1739.
- **Edge Betweenness (Table 7d, 9):** This method yielded the lowest fidelity, with Weighted Jaccard scores dropping to 0.1538. Similar to Label Propagation, it appears to have collapsed the majority of nodes into a single large component (C2.1) while stripping away smaller peripheral sets.

Finally, Figure 2 illustrates the topological layout of these communities. A detailed interpretation of why modularity optimization yields finer partitions than the sociological ground truth will be provided in the Discussion section.

Table 5: Global and Weighted Jaccard Scores for Synthetic Network (Reference: Ground Truth)

	J_{global}	J_w
Louvain	0.6765	0.1956
Label Propagation	0.7163	0.2500
Walktrap	0.7059	0.1769
Edge Betweenness	0.5588	0.1538

Table 6: Significance scores for Synthetic Network

	Louvain	Label Prop.	Walktrap	EdgeBetw	Ground Truth
Modularity	0.2958	0.0000	0.2213	0.1626	0.1666
Conductance	0.5759	0.0000	0.5257	0.7072	0.5748
Clustering Coef	0.1481	0.0859	0.0959	0.1789	0.1205
VI dist to GT	4.6909	1.9735	4.4862	5.2146	0.0000

Table 7: Matched Clusters and Local Jaccard Scores for Synthetic Network (Reference: Ground Truth)

a) Louvain vs Ground Truth			b) Label Propagation vs Ground Truth		
Ref (GT)	Match	Score	Ref (GT)	Match	Score
C1.1	C2.8	0.1739	C1.1	C2.1	0.3300
C1.2	C2.3	0.1538	C1.2	C2.1	0.2000
C1.3	C2.3	0.2308	C1.3	C2.1	0.2250
C1.4	C2.1	0.2239	C1.4	C2.1	0.2450

c) Walktrap vs Ground Truth			d) Edge Betweenness vs Ground Truth		
Ref (GT)	Match	Score	Ref (GT)	Match	Score
C1.1	C2.4	0.2143	C1.1	C2.1	0.1619
C1.2	C2.4	0.1957	C1.2	C2.1	0.1163
C1.3	C2.4	0.1500	C1.3	C2.1	0.2317
C1.4	C2.5	0.1475	C1.4	C2.1	0.1053

Table 8: Jaccard Similarity Matrices for Synthetic Network – Comparison of Algorithms

a) Louvain										
	C2.1	C2.2	C2.3	C2.4	C2.5	C2.6	C2.7	C2.8	C2.9	
C1.1	0.0879	0.0370	0.0521	0.1299	0.0759	0.0964	0.1392	0.1739	0.0411	
C1.2	0.0735	0.1154	0.1538	0.0339	0.0926	0.0656	0.0492	0.0377	0.0638	
C1.3	0.0685	0.0862	0.2308	0.1186	0.0492	0.0448	0.0952	0.0000	0.0185	
C1.4	0.2239	0.0635	0.0633	0.0294	0.0794	0.1563	0.0580	0.0159	0.0536	

b) Label Propagation		C2.1
C1.1		0.330
C1.2		0.200
C1.3		0.225
C1.4		0.245

c) Walktrap											
	C2.1	C2.2	C2.3	C2.4	C2.5	C2.6	C2.7	C2.8	C2.9	C2.10	C2.11
C1.1	0.0263	0.1282	0.0779	0.2143	0.0482	0.0395	0.0000	0.0135	0.0417	0.1159	0.0725
C1.2	0.0400	0.0690	0.0000	0.1957	0.0893	0.0392	0.0435	0.0652	0.0652	0.0200	0.0000
C1.3	0.0755	0.1356	0.1481	0.1500	0.0476	0.0175	0.0816	0.0189	0.0000	0.0182	0.0000
C1.4	0.0702	0.0000	0.0476	0.1226	0.1475	0.1273	0.0364	0.0741	0.0545	0.0169	0.0556

Table 9: Edge Betweenness Jaccard Similarity Matrix

	C2.1	C2.2	C2.3	C2.4	C2.5	C2.6	C2.7	C2.8	C2.9	C2.10
C1.1	0.1619	0	0.0149	0.0145	0	0	0.0455	0	0	0
C1.2	0.1163	0	0	0	0.0222	0	0	0.0244	0	0
C1.3	0.2317	0.0204	0	0.0208	0	0	0	0.0217	0	0.0217
C1.4	0.1053	0.0800	0.0200	0.0392	0.1000	0.0408	0	0	0.0204	0.0200

	C2.11	C2.12	C2.13	C2.14	C2.15	C2.16	C2.17	C2.18	C2.19	C2.20
C1.1	0.0149	0.0303	0.0147	0.0149	0.0298	0	0.0298	0.0149	0.0298	0.0303
C1.2	0.0243	0	0	0	0	0.0731	0.0238	0	0	0
C1.3	0	0	0.0435	0.0217	0	0	0	0	0	0
C1.4	0	0	0	0.0196	0.0192	0	0.02	0.0196	0	0

	C2.21	C2.22	C2.23	C2.24	C2.25	C2.26	C2.27	C2.28	C2.29	C2.30
C1.1	0.0147	0.0447	0.0294	0	0.0447	0.0149	0	0	0.0149	0.0147
C1.2	0.0238	0	0.0232	0.0250	0	0.0244	0	0.0500	0	0
C1.3	0.0212	0	0.0208	0	0	0	0.0217	0	0	0.0212
C1.4	0	0.0192	0	0	0.0192	0	0.0200	0	0.0196	0.0200

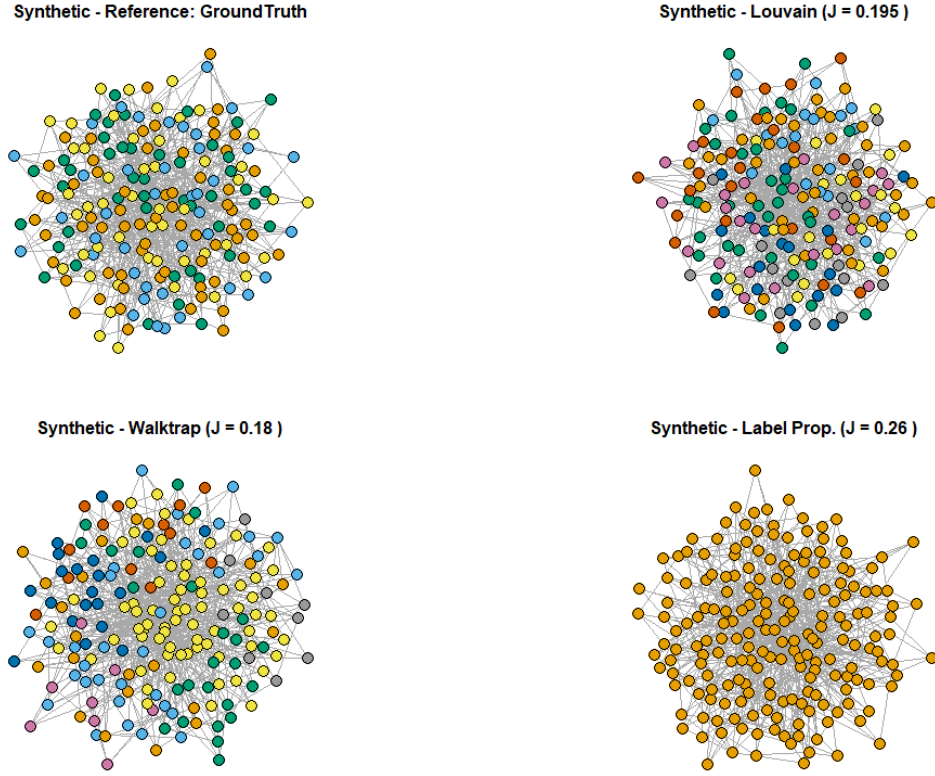


Figure 2: Synthetic network

2.3 Enron Network

The community detection results for the Enron email network are summarized in Tables 10 through 13 and visualized in Figure 3.

As established in the methodology, the **Louvain algorithm** was selected as the reference partition. This selection is supported by the significance scores presented in Table 11, where Louvain achieved a competitive modularity ($Q \approx 0.345$) and internal density compared to the alternative methods.

Table 10 presents the Global Weighted Mean Jaccard Scores (J_{global}) comparing the other algorithms against the Louvain reference. **Edge Betweenness** demonstrated the highest similarity to the reference structure ($J_{global} = 0.5575$), followed by Label Propagation and Walktrap.

The granular breakdown of these similarities is detailed in Table 12 (Matched Clusters) and Table 13 (Jaccard Similarity Matrices).

- **Label Propagation (Table 12b, 13b):** The data reveals a significant consolidation behavior, where clusters C1.1 through C1.7 from the reference partition are all matched to a single cluster (C2.1) in the Label Propagation result with low individual Jaccard scores (ranging from 0.08 to 0.18).
- **Walktrap (Table 12c, 13c):** This algorithm produces a more fragmented distribution. While it successfully identifies matches for certain communities (e.g., C1.7

matches C2.3 with score 0.9474), other reference clusters are split or matched with lower confidence.

- **Edge Betweenness (Table 12d, 13d):** This method shows the strongest localized matches, particularly for clusters C1.1 (0.8125), C1.7 (0.7895), and perfect matches for the smaller clusters C1.8 and C1.9.

Finally, Figure 3 illustrates the topological layout of these communities. A detailed interpretation of these topological differences and the algorithmic behaviors driving these results will be provided in the Discussion section.

Table 10: Global Weighted Mean Jaccard Scores for Enron Network

	J_{global}	J_w
Louvain (Reference)	1.0000	1.0000
Walktrap	0.1599	0.6726
Edge Betweenness	0.5575	0.4901
Label Propagation	0.2900	0.3333

Table 11: Significance Scores for ENRON Email Network

	Louvain	Label Prop.	Walktrap	Edge Betw.
Modularity	0.3448	0.0114	0.2951	0.1389
Conductance	0.4670	0.0119	0.4417	0.6963
Internal Density	0.4400	0.1437	0.4414	0.5762
Clustering Coef.	0.6126	0.3813	0.5985	0.6609

Table 12: Matched Clusters and Local Jaccard Scores for All Algorithms

a) Louvain (vs Ref)			b) Label Propagation		
Ref	Match	Score	Ref	Match	Score
C1.1	C2.1	1.0000	C1.1	C2.1	0.0879
C1.2	C2.2	1.0000	C1.2	C2.1	0.1484
C1.3	C2.3	1.0000	C1.3	C2.1	0.1813
C1.4	C2.4	1.0000	C1.4	C2.1	0.1813
C1.5	C2.5	1.0000	C1.5	C2.1	0.1593
C1.6	C2.6	1.0000	C1.6	C2.1	0.1374
C1.7	C2.7	1.0000	C1.7	C2.1	0.1044
C1.8	C2.8	1.0000	C1.8	C2.2	1.0000
C1.9	C2.9	1.0000	C1.9	C2.3	1.0000

c) Walktrap			d) Edge Betweenness		
Ref	Match	Score	Ref	Match	Score
C1.1	C2.5	0.6250	C1.1	C2.1	0.8125
C1.2	C2.7	0.5926	C1.2	C2.2	0.1528
C1.3	C2.2	0.4493	C1.3	C2.2	0.0854
C1.4	C2.2	0.4085	C1.4	C2.2	0.0595
C1.5	C2.1	0.6304	C1.5	C2.2	0.4167
C1.6	C2.6	0.4000	C1.6	C2.2	0.0946
C1.7	C2.3	0.9474	C1.7	C2.6	0.7895
C1.8	C2.11	1.0000	C1.8	C2.43	1.0000
C1.9	C2.13	1.0000	C1.9	C2.63	1.0000

Table 13: Jaccard Similarity Matrices

a) Louvain										
	C2.1	C2.2	C2.3	C2.4	C2.5	C2.6	C2.7	C2.8	C2.9	
C1.1	1.0	0	0	0	0	0	0	0	0	
C1.2	0	1.0	0	0	0	0	0	0	0	
C1.3	0	0	1.0	0	0	0	0	0	0	
C1.4	0	0	0	1.0	0	0	0	0	0	
C1.5	0	0	0	0	1.0	0	0	0	0	
C1.6	0	0	0	0	0	1.0	0	0	0	
C1.7	0	0	0	0	0	0	1.0	0	0	
C1.8	0	0	0	0	0	0	0	1.0	0	
C1.9	0	0	0	0	0	0	0	0	1.0	

b) Label Propagation										
	C2.1	C2.2	C2.3							
C1.1	0.088	0	0							
C1.2	0.148	0	0							
C1.3	0.181	0	0							
C1.4	0.181	0	0							
C1.5	0.159	0	0							
C1.6	0.137	0	0							
C1.7	0.104	0	0							
C1.8	0.000	1.0	0							
C1.9	0.000	0	1.0							

c) Walktrap														
	C2.1	C2.2	C2.3	C2.4	C2.5	C2.6	C2.7	C2.8	C2.9	C2.10	C2.11	C2.12	C2.13	C2.14
C1.1	0	0	0	0.22	0.63	0	0	0	0.13	0	0	0	0	0
C1.2	0.06	0.08	0	0	0	0	0.59	0	0	0	0	0	0	0
C1.3	0.03	0.45	0	0	0	0	0	0	0	0	0	0	0	0
C1.4	0.01	0.41	0	0.05	0	0	0	0	0	0	0	0	0	0.03
C1.5	0.63	0	0	0	0	0	0	0	0	0	0	0	0	0
C1.6	0.15	0	0	0	0	0.40	0	0.16	0	0.04	0	0.04	0	0
C1.7	0.02	0	0.95	0	0	0	0	0	0	0	0	0	0	0
C1.8	0	0	0	0	0	0	0	0	0	0	1.0	0	0	0
C1.9	0	0	0	0	0	0	0	0	0	0	0	0	1.0	0

d) Edge Betweenness (Selected)					
	C2.1	C2.2	C2.6	C2.43	C2.63
C1.1	0.81	0.01	0	0	0
C1.2	0	0.15	0	0	0
C1.3	0	0.09	0	0	0
C1.4	0	0.06	0	0	0
C1.5	0	0.42	0	0	0
C1.6	0	0.09	0	0	0
C1.7	0	0	0.79	0	0
C1.8	0	0	0	1.0	0
C1.9	0	0	0	0	1.0

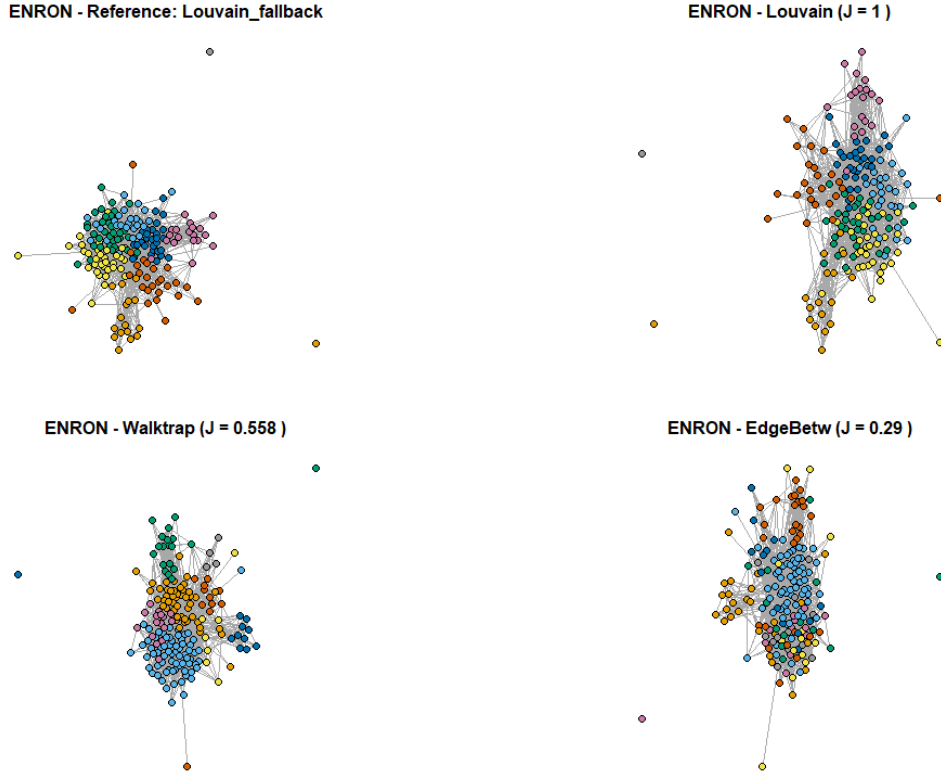


Figure 3: Ernon Network

2.4 Macaque Network

The community detection results for the Macaque cortical network are summarized in Tables 14 through 17 and visualized in Figure 4.

As established in the methodology, the **Louvain algorithm** was selected as the reference partition. This selection is supported by the significance scores presented in Table 15, where Louvain achieved the highest modularity ($Q \approx 0.3742$) among all methods and a strong internal density (0.6781).

Table 14 presents the Global Weighted Mean Jaccard Scores (J_{global}) comparing the other algorithms against the Louvain reference. **Walktrap** demonstrated the highest similarity to the reference structure ($J_{global} = 0.7511$), followed by Edge Betweenness and Label Propagation.

The granular breakdown of these similarities is detailed in Table 16 (Matched Clusters) and Table 17 (Jaccard Similarity Matrices).

- **Label Propagation (Table 16b, 17b):** The data reveals a merging behavior, where distinct reference clusters C1.1 and C1.2 are both mapped to the same target cluster (C2.1) with moderate similarity scores (0.50 and 0.44 respectively). In contrast, reference cluster C1.3 is well-preserved, matching C2.2 with a score of 0.7857.
- **Walktrap (Table 16c, 17c):** This algorithm aligns closely with the reference structure, achieving a perfect match for cluster C1.3 (1.00) and a high score for C1.1 (0.80).

However, divergence is observed in reference cluster C1.2, which is matched to C2.4 with a significantly lower score (0.5294), indicating some topological disagreement in that region.

- **Edge Betweenness (Table 16d, 17d):** This method shows strong localized performance for clusters C1.1 (0.8889) and C1.3 (0.8333). However, it struggles significantly with reference cluster C1.2, yielding the lowest best-match score among all algorithms (0.3529), suggesting a fragmentation of this community.

Finally, Figure 4 illustrates the topological layout of these communities. A detailed interpretation of these topological differences and the algorithmic behaviors driving these results will be provided in the Discussion section.

Table 14: Global Weighted Mean Jaccard Scores for Macaque Network

	J_{global}	J_w
Louvain (Reference)	1.0000	1.0000
Walktrap	0.7511	0.7765
Edge Betweenness	0.6716	0.6917
Label Propagation	0.5540	0.5756

Table 15: Significance Scores for Macaque Network

	Louvain	Label Prop.	Walktrap	Edge Betw.
Modularity	0.3742	0.1767	0.3213	0.2712
Conductance	0.3029	0.0946	0.2957	0.4556
Internal Density	0.6781	0.3872	0.6448	0.7534
Clustering Coef.	0.7042	0.5778	0.7255	0.7591

Table 16: Matched Clusters and Local Jaccard Scores for All Algorithms (Macaque Network)

a) Louvain (vs Ref)			b) Label Propagation		
Ref	Match	Score	Ref	Match	Score
C1.1	C2.1	1.0000	C1.1	C2.1	0.5000
C1.2	C2.2	1.0000	C1.2	C2.1	0.4412
C1.3	C2.3	1.0000	C1.3	C2.2	0.7857
c) Walktrap			d) Edge Betweenness		
Ref	Match	Score	Ref	Match	Score
C1.1	C2.2	0.8000	C1.1	C2.1	0.8889
C1.2	C2.4	0.5294	C1.2	C2.4	0.3529
C1.3	C2.1	1.0000	C1.3	C2.9	0.8333

Table 17: Jaccard Similarity Matrices - Macaque Network

a) Louvain				b) Label Propagation			
	C2.1	C2.2	C2.3		C2.1	C2.2	
C1.1	1.0	0	0	C1.1	0.50	0	
C1.2	0	1.0	0	C1.2	0.44	0.07	
C1.3	0	0	1.0	C1.3	0.02	0.79	

c) Walktrap					d) Edge Betweenness (Selected)				
	C2.1	C2.2	C2.3	C2.4		C2.1	C2.4	C2.9	C2.12
C1.1	0	0.80	0	0	C1.1	0.89	0	0	0
C1.2	0	0.12	0.24	0.53	C1.2	0.03	0.35	0	0.12
C1.3	1.0	0	0	0	C1.3	0.03	0	0.83	0

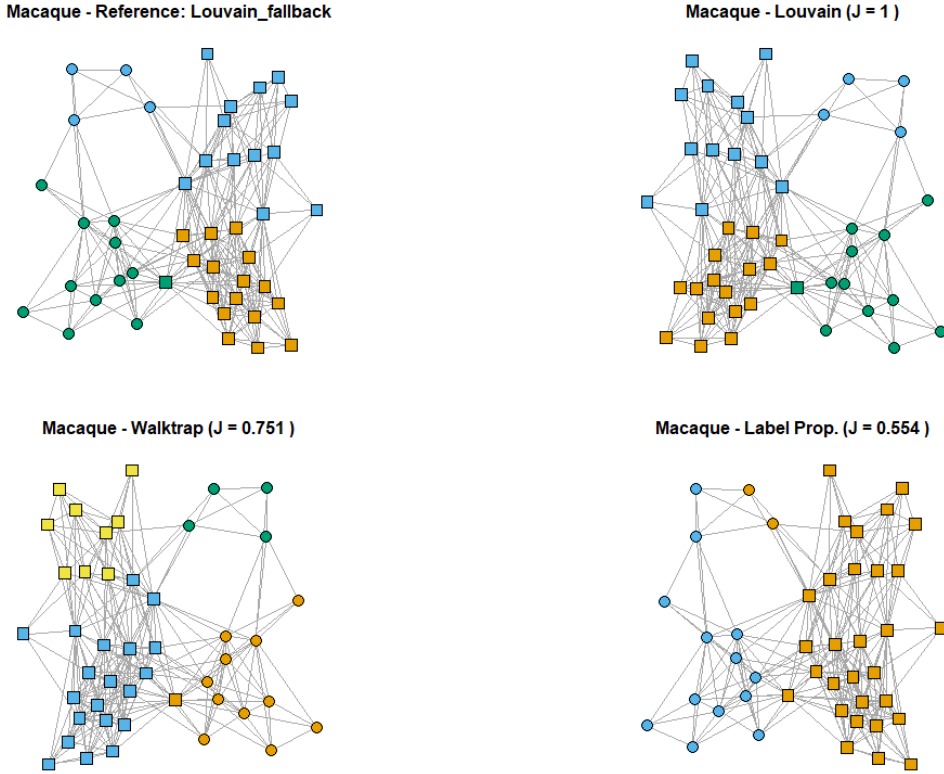


Figure 4: Macaque Network

3 Discussion

3.1 Discussion on Karate Network

3.1.1 Global and Weighted Similarity

The near-identical values between global and weighted Jaccard scores across all algorithms in Table 1 provide important insights into the nature of the detected partitions. The maxi-

imum deviation observed is only 0.0010 (for Walktrap: 0.7059 vs 0.7049), suggesting that all algorithms produced relatively balanced partitions where no single cluster dominates the network. This balance indicates that none of the methods created severely skewed community structures with, for example, one very large cluster and several tiny peripheral groups.

3.1.2 Significance Scores

The significance metrics in Table 2 reveal a striking paradox: algorithmic partitions (Walktrap $Q \approx 0.4112$, Louvain $Q \approx 0.4088$) achieve higher modularity than the actual social division ($Q \approx 0.3715$), yet the ground truth exhibits superior conductance (0.0952 vs. 0.2306–0.4194). This apparent contradiction provides fundamental insights about the relationship between mathematical optimization and sociological reality.

Modularity optimization seeks to maximize the difference between observed and expected intra-community edge density, favoring partitions with tightly-knit clusters relative to a random null model. The fact that Walktrap and Louvain exceed the ground truth’s modularity suggests they identified configurations with even denser internal connections than the real social split. However, this does not necessarily indicate superior community detection—it means they optimized a specific mathematical objective that may not perfectly align with how human communities actually form and dissolve.

The conductance discrepancy is particularly revealing: the ground truth’s substantially lower conductance (0.0952) indicates it achieves much cleaner separation between factions, with significantly fewer boundary edges. Algorithmic partitions, despite their higher modularity, exhibit two to four times more inter-community connections (conductance 0.17–0.42). This suggests that real social divisions are driven by factors beyond pure connectivity patterns such as personal relationships, leadership conflicts, resource competition, and shared activities created sharper boundaries in the actual club split than purely topological optimization could capture.

The Clustering Coefficient results show that Louvain (0.6025) and Walktrap (0.6138) approach the ground truth value (0.5341), indicating they successfully identify regions with high triadic closure. However, Edge Betweenness poor performance (0.4505) reflects its tendency to break apart cohesive triadic structures during the divisive process.

The Variation of Information (VI) results support these conclusions: Edge Betweenness performs very poorly ($VI = 1.6382$) because it splits the network into too many small communities. Louvain (0.7835) and Walktrap (0.8729), on the other hand, produce partitions that are much closer to the ground truth, even though their modularity is higher.

3.1.3 Cluster Scores

The cluster-level analysis in Tables 3 and 4 reveals distinct algorithmic behaviors and their structural implications.

Despite its algorithmic simplicity, Label Propagation achieved the highest global similar-

ity (0.7163) and demonstrated notably strong recovery of the “Officers” faction (Jaccard = 0.7778). The similarity matrix (Table 4b) shows near-zero cross-contamination, suggesting it successfully captured the binary split without over-partitioning. The algorithm correctly identified the two-community structure with minimal confusion between factions. However, the slight overlap (0.0345 between C1.2 and C2.1) reflects its stochastic nature because random tie-breaking during label propagation can shift boundary nodes between communities across different runs.

Both Louvain and Walktrap demonstrated a clear pattern of subdividing ground truth clusters. For instance, Louvain split reference cluster C1.1 into multiple subgroups (C2.1 and C2.2), reducing the primary match score to 0.6875 despite achieving higher overall modularity. This behavior reflects the well-known resolution limit problem in modularity optimization size scales and may subdivide or merge communities to maximize Q , even when a coarser partition better reflects social reality. The Karate Club’s small size (34 nodes) places it near this resolution threshold, causing modularity methods to detect substructure within the two main factions that may or may not correspond to meaningful social subdivisions.

The over-partitioning into six communities (Tables 3d, 4d) with scattered Jaccard similarities across the matrix reveals the fundamental challenge of divisive hierarchical methods: determining the optimal cut level in the dendrogram. Edge Betweenness successfully identifies bridge edges (as evidenced by its ability to progressively separate the network), but without an objective stopping criterion, it continues subdividing beyond the natural two-community structure. For this reason, the resulting fragmentation explains its poor performance across all global metrics.

3.2 Discussion on Synthetic Network

The analysis of the synthetic network reveals how different community-detection algorithms behave when the underlying structure is relatively weak and the communities are only moderately separated. In this network, the intra-community edge probability is set to 0.7 and the inter-community probability to 0.3, meaning that communities are present but not strongly insulated from each other. Moreover, the Barabási-Albert growth within blocks produces heterogeneous degrees and hub nodes, which tend to blur the separation between communities. These structural properties are reflected in the results obtained across all evaluation metrics.

3.2.1 Global and Weighted Similarity

A first observation comes from the comparison between global and weighted Jaccard similarity scores. Global similarity values appear relatively high suggesting that the algorithms are capable of identifying subsets of nodes that overlap with the ground-truth communities. However, the weighted Jaccard scores in all the algorithms drop to very low values (Louvain

0.1956, Walktrap 0.1769, Edge Betweenness 0.1538). This discrepancy indicates substantial heterogeneity in community sizes and uneven recovery of the four planted communities.

According to the theory, weighted similarity provides a more faithful assessment when cluster sizes vary, because each community contributes proportionally to its size. The large drop from global to weighted Jaccard therefore reflects that algorithms tend to recover certain communities well while performing poorly on smaller or noisier blocks. Given the relatively weak separation in the construction of the network (0.7 vs. 0.3), this outcome is expected: high-degree hubs and cross-block edges produce overlap that algorithms cannot resolve cleanly, leading to partial matches rather than full recovery of all four communities.

3.2.2 Significance Scores

The significance metrics further highlight how the weakly modular structure of the network influences algorithmic performance. Modularity is low across all methods (Louvain 0.2958, Walktrap 0.2213), and even the ground truth yields only 0.1666. From theory, modularity measures the excess of internal edges relative to the configuration null model; low modularity indicates that internal density is not much higher than what could arise randomly from the degree distribution. This is consistent with the generative model: with inter-community probabilities as high as 0.3, the frontier between communities is relatively porous, and hubs connect across blocks, reducing the modular structure.

The conductance scores also confirm the lack of clear boundaries. Good communities should minimize external connectivity (low conductance), but all algorithms report extremely high values (between 0.52 and 0.70), roughly similar to or worse than the ground truth. This aligns with the theoretical definition: conductance depends on the size of the cut between communities, and in this synthetic network many edges cross community borders, so no method can produce well-separated partitions.

The clustering coefficient, a measure of internal triadic closure, is also low for all partitions. This reflects the fact that the stochastic block model with Barabási–Albert internal wiring does not produce many triangles compared to real social networks. Edge Betweenness displays the highest coefficient (0.1789), but still far from dense triadic structure, confirming that communities are not strongly cohesive.

Finally, the Variation of Information (VI) scores illustrate that all algorithms deviate substantially from the ground truth. Louvain and Walktrap produce the closest partitions, but their VI distances (4.69 and 4.48) are still very high, while Edge Betweenness performs the worst (5.21). In theory, VI quantifies the information needed to transform one partition into another; large values here indicate that algorithms either merge communities incorrectly or split them into too many subgroups, which is consistent with the weak structural separability in the network.

3.2.3 Cluster Scores

The cluster score analysis reinforces the observation that these algorithms struggle to align with the four planted communities at the level of individual blocks. Across all methods, the weighted Jaccard indices remain consistently low, ranging from 0.154 (Edge Betweenness) to a maximum of 0.25 (Label Propagation). This indicates that none of the methods fully recover the ground truth; instead, they identify high-quality matches only for specific subsets of nodes while misinterpreting the broader structure. This divergence manifests primarily as a tendency toward over-aggregation in the density-based and random-walk methods.

Label Propagation illustrates this most clearly: despite achieving the highest weighted mean, it collapses the distinct four-block structure into a single dense macro-region (C2.1). This outcome confirms the method’s known sensitivity to degree heterogeneity, where a single core absorbs all ground-truth clusters.

Walktrap exhibits a similar, though slightly more segmented, pattern of aggregation. By merging three of the four planted communities into one cluster (C2.4) and isolating only the fourth, Walktrap demonstrates a preference for maximizing local modularity over preserving the planted partition. Louvain suffers from a related issue driven by the resolution limit of modularity optimization. While it avoids the total collapse seen in Label Propagation, it nevertheless re-aggregates weak or small communities to maximize modularity gain. Its best correspondences map ground-truth clusters to different estimated blocks with only moderate Jaccard values (0.15–0.23), reflecting a structural divergence where the algorithm prioritizes mathematical optimization over the intended social reality of the planted blocks. In sharp contrast to these aggregation tendencies, Edge Betweenness fails through excessive fragmentation. Although the algorithm generates a high quantity of clusters (over 60), the best match for all four ground-truth blocks paradoxically maps to the same single estimated cluster. This confirms that while the removal of high-betweenness edges successfully shatters the network’s periphery into small, noisy components, it fails to disentangle the core structure, leaving the most central nodes concentrated in a single block that captures only fragments of the true communities.

3.3 Discussion on Enron Network

3.3.1 Justification of the Reference Partition

The choice to use the Louvain partition as ground truth comes from the distinction between significance and stability.

Significance measures, like Modularity (Q), rely on a specific null model - typically the configuration model where edges are randomized while preserving degree distribution.

Optimizing Q simply maximizes the deviation from this random noise. However, a partition can be statistically significant (high Q) without necessarily being the best representation of the network’s organization. By selecting Louvain as a reference and evaluating how other algorithms align with it (via Jaccard similarity), we are effectively assessing the stability of

the community structure.

3.3.2 Global and Weighted Similarity

The results indicate that **Edge Betweenness** achieved the highest global similarity to the reference ($J_{global} = 0.5575$).

The high agreement with Louvain suggests that the Enron network possesses well-defined bottlenecks or bridge edges connecting distinct departments. In contrast, Label Propagation performed poorly ($J_{global} = 0.29$). Since Label Propagation relies on the rapid diffusion of labels, its failure (collapsing the network into a giant component) implies that the network’s core is too dense for simple diffusion to identify boundaries, requiring the more aggressive approach of Edge Betweenness or the local optimization of Louvain.

3.3.3 Significance Scores

The significance scores highlight the trade-offs between different quality definitions. Louvain achieved the highest modularity (0.3448), which is expected as it explicitly optimizes the difference between observed intra-cluster edges and the expected number in a random graph. In contrast, Edge Betweenness showed a high internal density (0.5762), a result consistent with the algorithm’s divisive nature. By severing connections between groups, it leaves behind highly connected cliques or dense cores. On the contrary, the poor performance of Label Propagation resulted in a conductance score near zero for the giant component, failing the criterion that communities should be sparse with respect to the outside.

3.3.4 Cluster Scores

The analysis confirms the giant component issue in Label Propagation, where multiple reference clusters were merged. Conversely, Edge Betweenness showed near-perfect matches for smaller, specific reference clusters. This indicates that while Edge Betweenness may over-divide the network, the resulting fragments are highly pure subsets of the larger communities detected by Louvain.

3.4 Discussion on Macaque Network

3.4.1 Justification of the Reference Partition

The selection of the Louvain partition as the reference is based on it achieving the highest modularity score ($Q \approx 0.3742$) among all tested algorithms. Modularity measures the difference between the number of edges falling within groups and the expected number in an equivalent network with edges placed at random. However, relying solely on modularity is insufficient for validation, as high Q values can sometimes result from the algorithm’s specific resolution bias rather than the true scale of the network’s structure. Therefore, we utilize Jaccard similarity to assess the stability of the partition. Stability implies that meaningful

clusters should capture an inherent structure in the data and not be sensitive to the particularities of the clustering algorithm. The high agreement between Walktrap and Louvain validates the Louvain partition as a robust reference rather than a random local optimum.

3.4.2 Global and Weighted Similarity

The results indicate that **Walktrap** achieved the highest global similarity to the reference ($J_{global} = 0.7511$). This agreement is significant because both Louvain and Walktrap operate on hierarchical principles. Walktrap is an agglomerative hierarchical algorithm that joins clusters based on random walk similarity, while Louvain aggregates nodes into communities in passes. The convergence of these two methods suggests that the Macaque cortical network possesses a strong, multi-level hierarchical organization. In contrast, Label Propagation ($J_{global} = 0.5540$) performed the worst, suggesting that the functional modules of the brain are not easily resolved by simple label diffusion processes, likely due to complex inter-module connectivity.

3.4.3 Significance Scores

The significance scores reveal distinct topological characteristics identified by each algorithm. Louvain achieved the highest modularity (0.3742), effectively maximizing the density of connections within groups relative to the configuration null model. Edge Betweenness achieved the highest Internal Density (0.7534). This metric, defined as the fraction of edges inside the cluster relative to the maximum possible number of internal edges, indicates that Edge Betweenness successfully isolated the most tightly connected cores or cliques of the brain network. This behavior is consistent with the algorithm’s design, which iteratively removes bridge edges with high betweenness. By cutting these bridges, the algorithm isolates dense subgraphs, resulting in high internal density but potentially over-fragmenting the broader community structure, as reflected in its lower global Jaccard score compared to Walktrap.

3.4.4 Cluster Scores

The analysis reinforces the finding that certain brain regions form highly distinct modules while others are more ambiguous. Reference cluster C1.3 was identified with high precision by all algorithms (Walktrap: 1.0, Edge Betweenness: 0.83), suggesting it represents a well-separated functional module with high internal connectivity. Conversely, Reference Cluster C1.2 proved difficult to resolve, with Edge Betweenness achieving a match score of only 0.35. This fragmentation suggests that C1.2 likely contains multiple bridge edges that Edge Betweenness severed, thereby splitting a single functional community into smaller components.

3.5 General Discussions

3.5.1 Jaccard Similarity vs. Significance Metrics

A critical methodological conclusion from this study is the preference for Jaccard-based stability analysis over internal significance metrics when ranking the quality of a partition relative to a reference.

Internal metrics like Modularity are absolute values based on a theoretical null model. A partition can have a positive Q simply because the network is not random, not because the specific partition is the correct sociological or biological division. Furthermore, Q is subject to resolution limits. In contrast, Jaccard Similarity measures the consensus between methods. As defined in stability analysis, meaningful clusters should remain unchanged under variations (such as changing the algorithm). Therefore, a high Jaccard score represents a cross-verification of the structure, providing a stronger guarantee of validity than a high Modularity score alone.

3.5.2 Weighted Scoring and Aggregation Strategies

We utilized a weighted average for global Jaccard scores, weighting by cluster size. This approach aligns with the standard practice for evaluating network-wide metrics, where the contribution of a community is proportional to its volume relative to the total network size (n_c/n).

Unweighted means can be heavily skewed by the dust of small, trivial clusters often produced by hierarchical algorithms at the bottom of the dendrogram. By weighting the Jaccard scores, we ensure that our evaluation focuses on the algorithm’s ability to recover the significant, structural backbones of the network rather than its handling of outliers.

The weighted approach is fundamentally more reasonable because it accounts for cluster size heterogeneity. In community detection, partitions often contain clusters of vastly different sizes because sometimes they start from small groups of 2-3 nodes to large core communities containing dozens of members. A simple mean treats each cluster’s Jaccard score equally, regardless of whether it represents 5% or 50% of the network. This can be misleading: a high-quality partition of a major community should contribute more to the overall similarity assessment than a trivial match of a tiny outlier cluster. The weighted mean, using node fraction as weights, ensures that larger, structurally significant communities are properly weighted in the final score, providing a truer representation of overall partition quality.

An alternative aggregation approach could use the median Jaccard score across all matched cluster pairs. Unlike the mean, the median is robust to outliers and would provide a measure of typical cluster matching quality without being skewed by either exceptionally good or poor individual cluster matches. This could be particularly useful when partitions contain a few anomalous tiny clusters or when comparing algorithms that produce different numbers of communities. However, the near-identical global and weighted scores in our Karate

Club results suggest relatively balanced cluster sizes across all algorithms, indicating none produced severely skewed partitions.

4 Methods

In this study, we implemented a comparative analysis of community detection algorithms using the R programming environment. The simulations and evaluations relied primarily on the `igraph` package for network manipulation and the `clustAnalytics` package for stability and significance assessment .

4.1 Cluster Similarity Quantification

To evaluate the quality of detected communities against a reference partition, we developed a custom evaluation metric based on the Jaccard index. While the standard Jaccard index $J(A, B) = \frac{|A \cap B|}{|A \cup B|}$ measures the similarity between two specific sets , we extended this to compare entire graph partitions through the following three-step procedure:

1. **Pairwise Similarity Matrix:** We implemented a function to compute a similarity matrix M where M_{ij} represents the Jaccard index between the i -th cluster of the reference partition and the j -th cluster of the detected partition.
2. **Best-Match Mapping:** To identify the closest correspondence between partitions, we implemented a matching function that, for each cluster in the reference set, identifies the cluster in the comparison set that maximizes the Jaccard index .
3. **Global Weighted Similarity:** We defined the global similarity score $J_{weighted}$ as the weighted mean of the best-match Jaccard indices. The weights were assigned proportional to the size of the clusters in the reference partition to ensure that larger communities contribute more to the overall score .

4.2 Evaluation Pipeline

We assessed four distinct community detection algorithms: `Louvain`, `Label Propagation`, `Walktrap`, and `Edge Betweenness`. To account for the stochastic nature of algorithms like `Walktrap` and `Label Propagation`, we designed an evaluation wrapper that performs the following steps for each network:

- **Repetitions:** The algorithms were executed over $N = 5$ independent runs.
- **Significance Scoring:** We utilized the `evaluate_significance` function to compute modularity, conductance, and other internal quality metrics, averaging these scores across the runs .

- **Reference Selection:** The reference clustering for the Jaccard comparison was determined dynamically. If a ground truth was available, it was used; otherwise, the algorithm yielding the highest average modularity score was selected as the reference.

4.3 Datasets and Preprocessing

The evaluation was performed on four networks with distinct topological characteristics:

- **Zachary’s Karate Club:** Used as a benchmark with a known ground truth of two factions.
- **Synthetic Scale-Free Network:** We generated a network with $n = 200$ nodes and approx. 800 edges using the Barabási-Albert block model (`barabasi_albert_blocks`). The topology was constrained to form 4 distinct communities.
- **ENRON Email Network:** The original multigraph was preprocessed into a simple, undirected, weighted graph. Multiple edges between users were aggregated by summing their counts to define edge weights.
- **Macaque Network:** We selected the Macaque cortical connectivity dataset from `igraphdata` as the network of choice with no ground truth. This graph was simplified to remove loops and multiple edges prior to analysis.